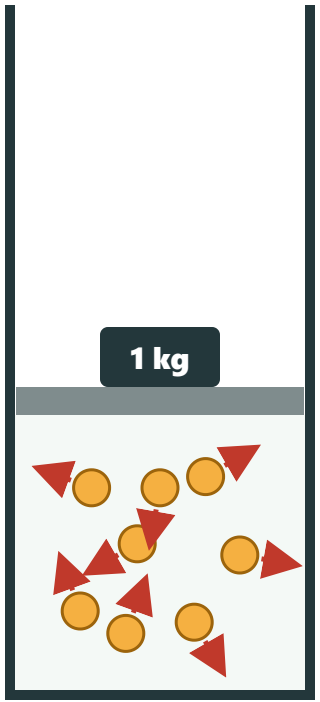
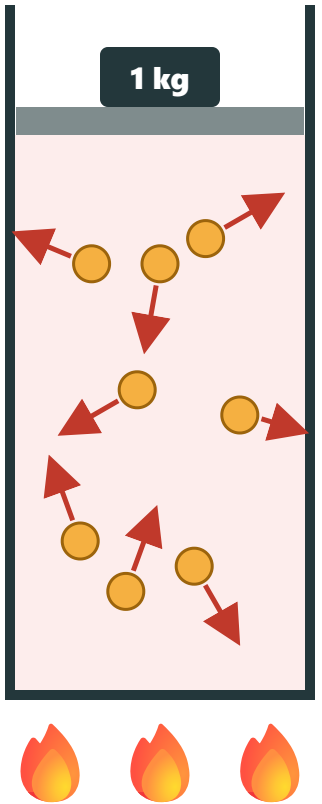
Charles' law: the pressure does not change



we heat it
T × 2



T = 300 K
V = 1 L

T = 600 K
V = 2 L

p constant ⚖️ (the piston moves freely)

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

Example: $\frac{1\ L}{300\ K} = \frac{2\ L}{600\ K}$

🎈 A balloon left in the sun **gets bigger**; in the freezer it **shrinks**.

🔥 In a hot-air balloon, the heated air **expands**, becomes less dense and the balloon rises.